General comments on EE2-21 Feedback Systems paper 2019

Overall, the students have done well on the exam paper, scoring an exam average of 72%.

1. The students have done relatively well on this question, scoring approximately 79%. This is an op–amp modeling, including state–space, proportional controller design and Nyquist stability analysis question and is a variation on a past exam question and typical study group questions.
   (a) Tests understanding of transfer functions for simple circuits involving op–amps.
   (b) Tests obtaining differential equations from given transfer functions.
   (c) Tests obtaining state–space realizations from transfer functions.
   (d) Tests understanding of the use of the final value theorem in simple analysis.
   (e) Tests understanding of the use of the final value theorem in simple design for second order systems.
   (f) Tests understanding of the use of the final value theorem in more complex design.
   (g) Tests sketching Nyquist diagrams for simple systems.
   (h) Tests the use of the Nyquist theorem in simple proportional stability design.

2. This is a Nyquist diagram analysis and design question that involves a a third order system. It also tests understanding of the small gain theorem. The students did less well on this question scoring approximately 57%.
   (a) Tests the students' understanding of the use of the Routh–Hurwitz stability criterion in sketching Nyquist diagrams.
   (b) Requires an understanding of the use of the Routh–Hurwitz stability criterion to determine the stability margins and the cross–over frequency.
   (c) Qualitative analysis of the effects of PI compensation on the performance.
   (d) Tests the students' understanding of the relation between Nyquist analysis and the small gain theorem. Also, tests the student's understanding of the use of the small gain theorem in robust stability analysis.

3. This is a question about using pole placement and the characteristic equation for designing a non–standard feedback loop. The students did very well on this question, scoring approximately 76%.
   (a) Tests the understanding of the use of the characteristic equation (CE) in pole placement design using one parameter. It also requires an understanding of the relation between pole location and design specifications.
   (b) Tests the understanding of the use of the characteristic equation (CE) in pole placement design using two parameter. It also requires an understanding of the relation between pole location and design specifications.